

NATTAVAARA, Sweden

WMO: 021610

Lat: 66.752N

Lon: 20.928E

Elev: 327

StdP: 97.46

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.4	-30.1	-37.6	0.1	-33.4	-33.9	0.2	-29.9	6.4	-4.6	5.7	-4.7	0.3	290	0.695

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	24.0	15.3	22.0	14.5	20.1	13.8	17.1	21.9	15.8	19.9	14.7	18.5	2.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	11.2	18.4	14.1	10.4	17.1	13.0	9.7	16.1	49.4	22.0	45.3	20.1	42.1	18.6	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 6.3	5.4	4.6		-37.2	26.7	3.2	3.4	-39.5	29.1	-41.3	31.1	-43.1	33.0	-45.4	35.4	
(5)			DB	-36.3	18.8	3.2	1.6	-38.6	19.9	-40.5	20.9	-42.3	21.8	-44.7	22.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-13.5	-12.9	-8.0	-1.7	5.3	11.0	13.8	11.6	6.4	-1.0	-7.2	-10.7
	DBStd	10.95	7.66	7.66	6.15	4.35	4.05	3.67	2.81	3.26	3.38	4.73	6.14	7.09
	HDD10.0	4100	729	642	558	350	155	31	3	20	114	340	516	642
	HDD18.3	6875	988	876	816	600	403	222	142	208	357	598	766	900
	CDD10.0	271	0	0	0	0	10	61	122	71	8	0	0	0
	CDD18.3	5	0	0	0	0	0	2	3	1	0	0	0	0
	CDH23.3	95	0	0	0	0	5	29	51	10	0	0	0	0
	CDH26.7	15	0	0	0	0	0	6	9	0	0	0	0	0
(14) Wind	WSAvg	1.9	1.4	1.5	2.1	2.1	2.3	2.5	2.2	1.9	1.9	1.8	1.4	1.4
(15) Precipitation	PrecAvg	597	42	31	31	28	44	66	89	73	53	48	49	44
	PrecMax	823	81	59	62	58	86	107	125	136	196	92	102	145
	PrecMin	408	4	6	9	8	10	15	30	27	14	16	15	10
	PrecStd	106	17	16	14	14	20	26	27	33	39	20	25	28
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	5.0	7.0	14.0	23.7	26.8	27.7	24.8	17.9	10.4	5.9	3.8
		MCWB	2.5	2.4	3.2	6.6	12.4	16.9	17.9	16.8	12.5	7.6	4.1	2.1
	2%	DB	0.6	2.5	4.6	10.0	20.1	23.0	24.5	22.3	15.7	8.3	3.7	1.2
		MCWB	-0.4	0.4	1.5	4.7	11.7	13.3	16.0	15.8	11.5	6.7	2.8	0.1
	5%	DB	-1.3	0.0	2.9	7.7	16.3	20.6	22.3	20.2	13.9	6.8	1.5	-0.1
		MCWB	-1.7	-1.1	0.4	3.6	9.7	12.8	15.2	14.9	10.8	5.4	0.9	-0.5
	10%	DB	-3.3	-1.9	1.1	5.6	13.1	18.5	20.3	18.0	12.2	5.2	0.2	-1.2
		MCWB	-3.7	-2.4	-0.7	2.3	7.7	11.9	14.4	13.7	9.7	4.2	-0.1	-1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.6	3.2	7.0	14.0	17.7	19.2	18.2	13.7	8.4	4.6	2.2
		MCDB	4.2	4.7	6.4	11.7	21.2	24.1	22.9	22.5	16.1	9.5	5.2	3.7
	2%	WB	-0.2	0.7	1.8	5.3	12.2	15.3	17.5	16.7	12.4	6.9	2.7	0.3
		MCDB	0.3	2.1	4.1	9.5	19.0	20.2	22.7	20.7	15.0	8.3	3.7	0.8
	5%	WB	-1.7	-1.1	0.7	3.8	10.1	13.7	16.2	15.4	11.2	5.6	1.0	-0.5
		MCDB	-1.2	-0.2	2.6	7.4	15.9	18.4	20.5	19.2	13.1	6.7	1.5	-0.1
	10%	WB	-3.7	-2.5	-0.5	2.5	8.4	12.5	15.1	14.2	10.2	4.3	0.0	-1.6
		MCDB	-3.4	-1.8	0.9	5.4	12.5	17.3	19.0	17.1	11.9	5.1	0.2	-1.2
(35) Mean Daily Temperature Range	MDBR		9.1	10.6	12.2	10.7	9.8	9.5	9.9	9.7	8.7	6.7	7.3	8.9
	5% DB	MCDBR	8.6	9.4	12.0	13.1	16.2	13.7	14.2	13.6	11.6	8.2	6.2	7.7
		MCWBR	8.2	8.5	9.4	8.4	8.6	6.3	6.9	7.5	7.6	6.6	5.7	7.4
	5% WB	MCDBR	8.1	9.0	11.0	11.8	15.1	11.8	11.7	11.8	9.7	7.7	5.8	7.6
		MCWBR	7.8	8.2	8.9	7.5	8.4	6.1	6.6	6.9	6.7	6.3	5.4	7.3
(40) Clear-Sky Solar Irradiance	taub		0.182	0.247	0.282	0.312	0.330	0.333	0.342	0.327	0.297	0.273	0.341	N/A
	taud		1.938	2.146	2.226	2.280	2.467	2.496	2.483	2.510	2.572	2.395	2.772	N/A
	Ebn at Noon		214	603	775	846	873	876	854	832	783	590	243	0
	Edh at Noon		14	50	79	98	91	91	90	79	58	40	16	0
(44) All-Sky Solar Radiation	RadAvg		0.07	0.61	2.11	4.00	4.95	5.14	4.79	3.53	1.97	0.70	0.14	0.00
	RadStd		0.01	0.08	0.14	0.18	0.48	0.40	0.47	0.31	0.28	0.11	0.02	0.00

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page